

Posets and Lattices

Math 737 - Lab 2

Sage includes many pre-defined posets. Let's explore these. Type posets with a . after it. Then press tab. You should see a list of built-in posets.

Exercise 1: Plot several built-in posets. You may need to insert a parameter such as 3 inside the parentheses at the end.

```
In [8]: B = posets.BubblePoset(3,2)
        H = posets.HochschildLattice(5)
        S = posets.SymmetricGroupWeakOrderPoset(4)
```

```
In [27]: g1 = B.plot(
        axes=False,
        )

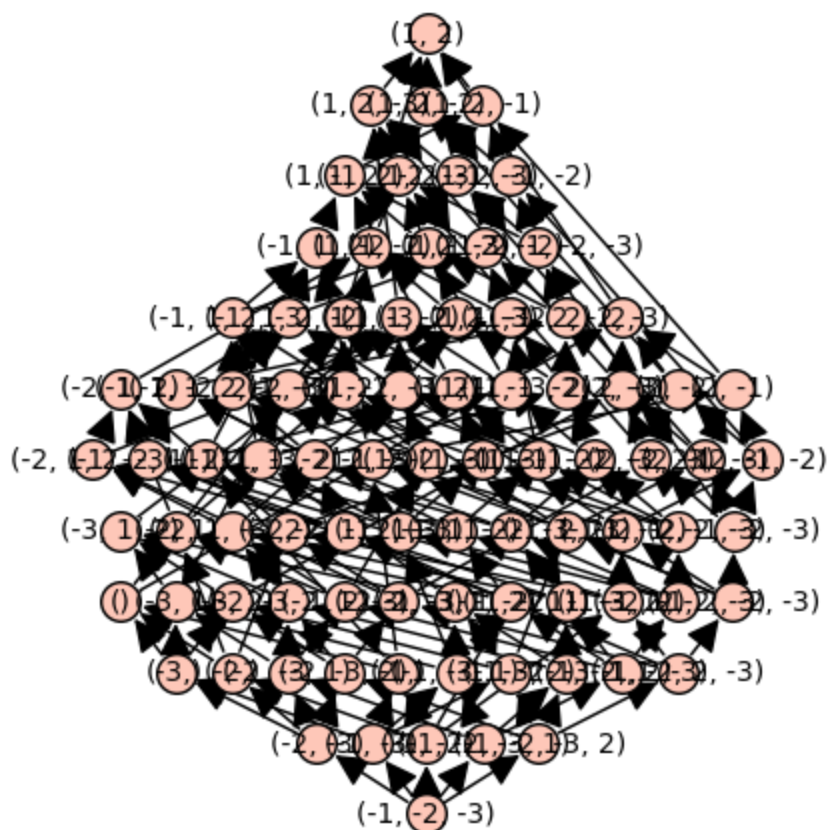
        g2 = H.plot(
        axes=False,
        )

        g3 = S.plot(
        axes=False,
        )

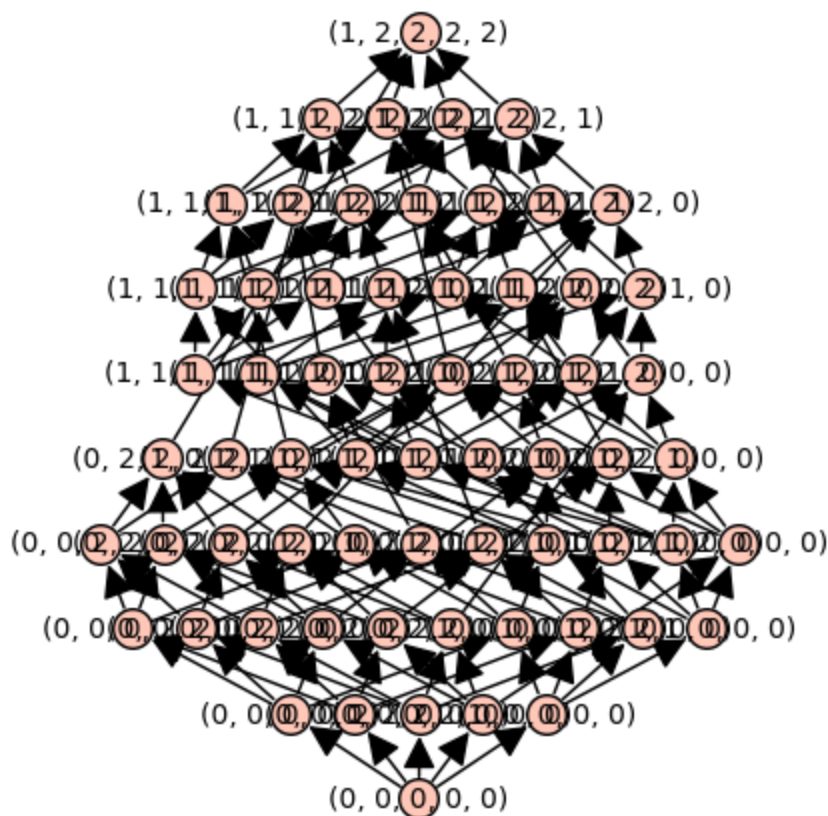
        h = B.height() # This is a constant to scale things
        g1.set_axes_range(-h, h, -1, h + 0.5) # makes some axes

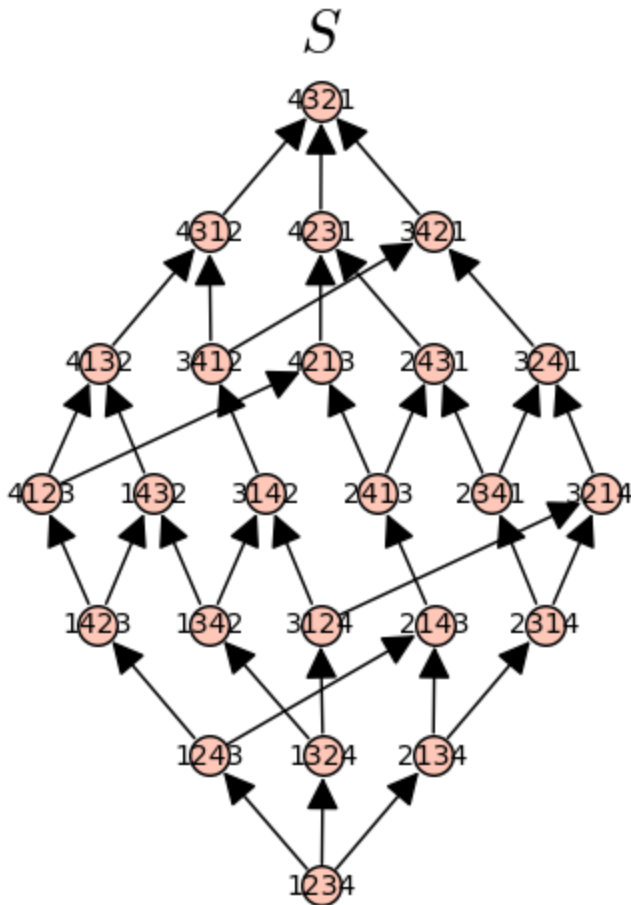
        (g1 + text(r"$B$", (5.5, 12),
        fontsize=22, color='black',
        horizontal_alignment='center')).show()
        (g2 + text(r"$H$", (4.5, 10),
        fontsize=22, color='black',
        horizontal_alignment='center')).show()
        (g3 + text(r"$S$", (3, 6.5),
        fontsize=22, color='black',
        horizontal_alignment='center')).show()
```

B



H





Sage includes many common combinatorial objects. Let's use Sage to make natural posets on combinatorial objects.

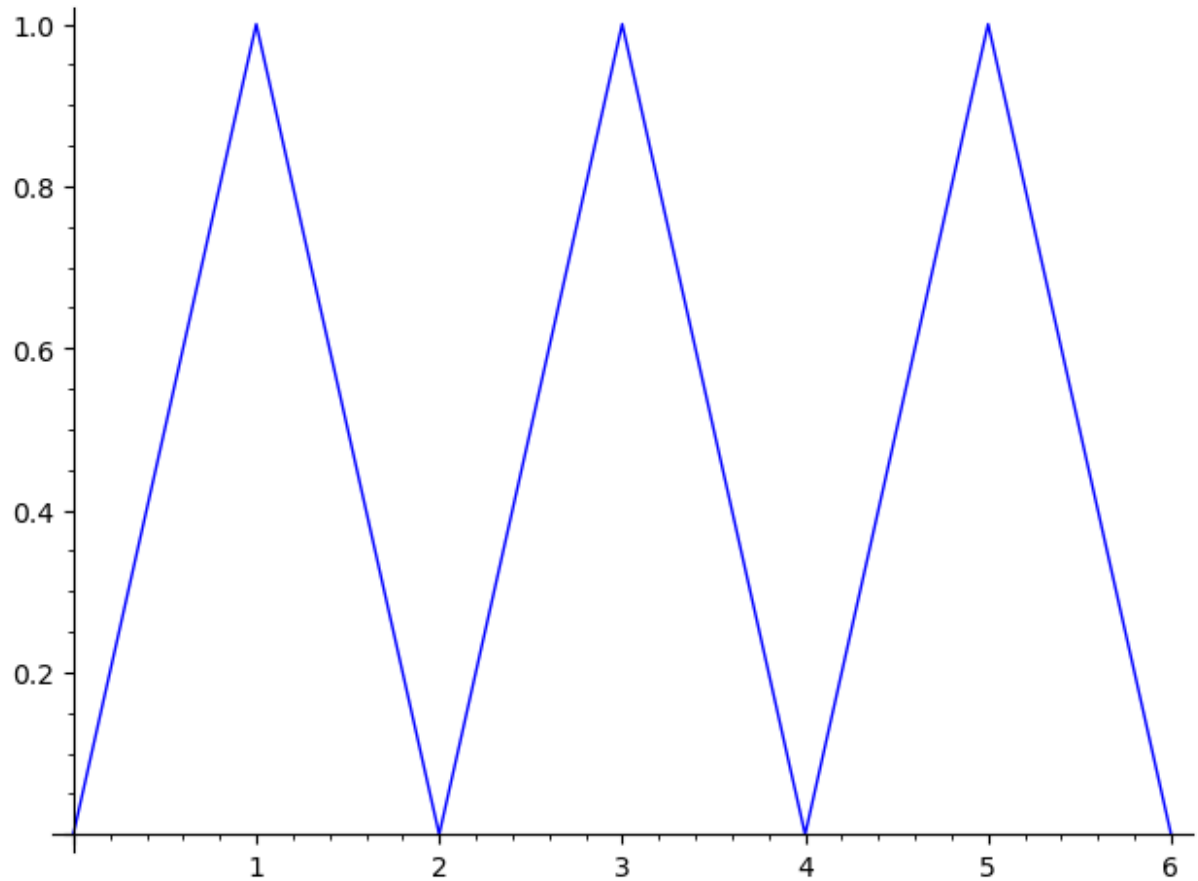
Until now, we have defined posets by inputting elements and covering relations. Now let's define a poset by specifying a set of combinatorial objects and the partial order as a function which returns 'True' if $x < y$ and 'False' otherwise.

Exercise 2: Evaluate the blocks and follow the directions below to define the poset on Dyck paths for $n=3,4,5$ with partial order of containment.

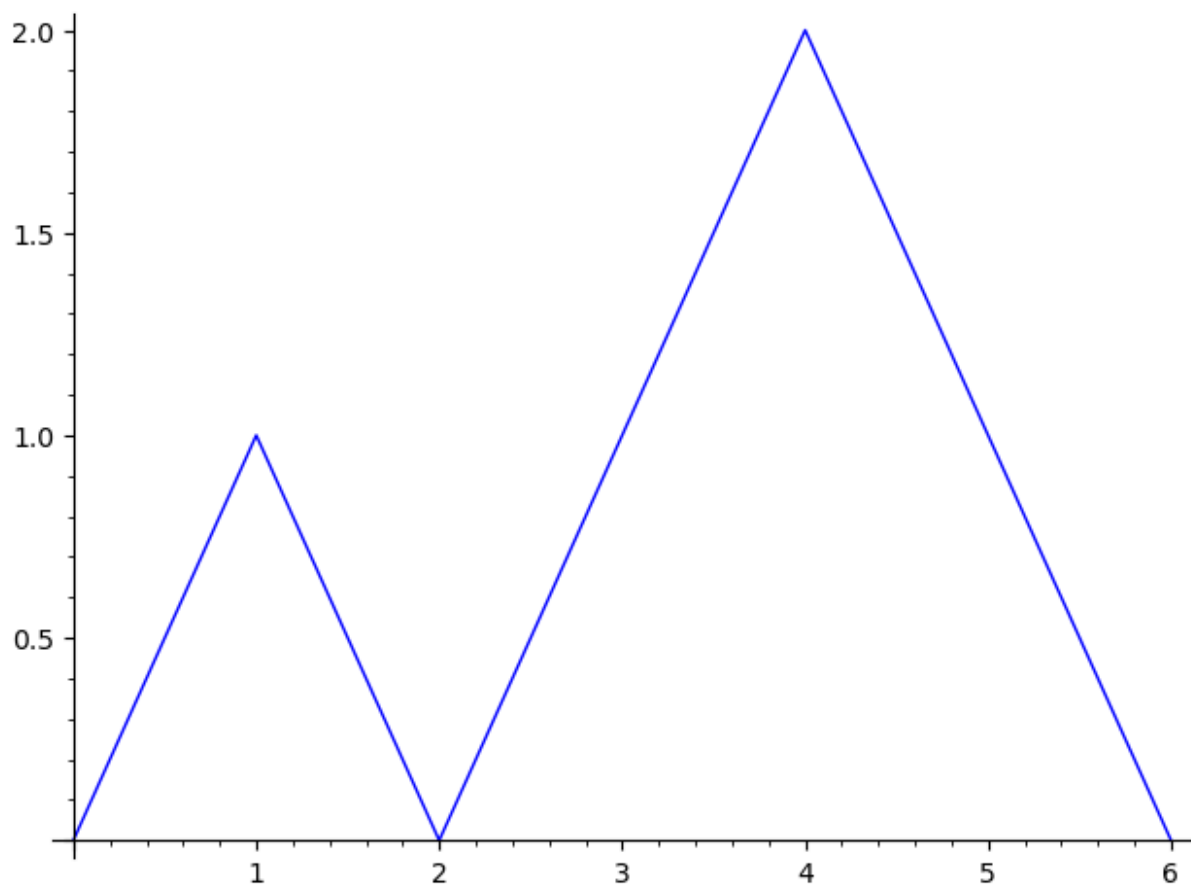
```
In [32]: DW = {
    "3": DyckWords(3),
    "4": DyckWords(4),
    "5": DyckWords(5)
}
```

```
In [36]: #for key,D in DW.items():
          #print(f"Dyck words with {key} pairs of parantheses:")
          for d in DW["3"]:
              print(d)
              (d.plot()).show()
              d.to_partition()
              print("-----")
```

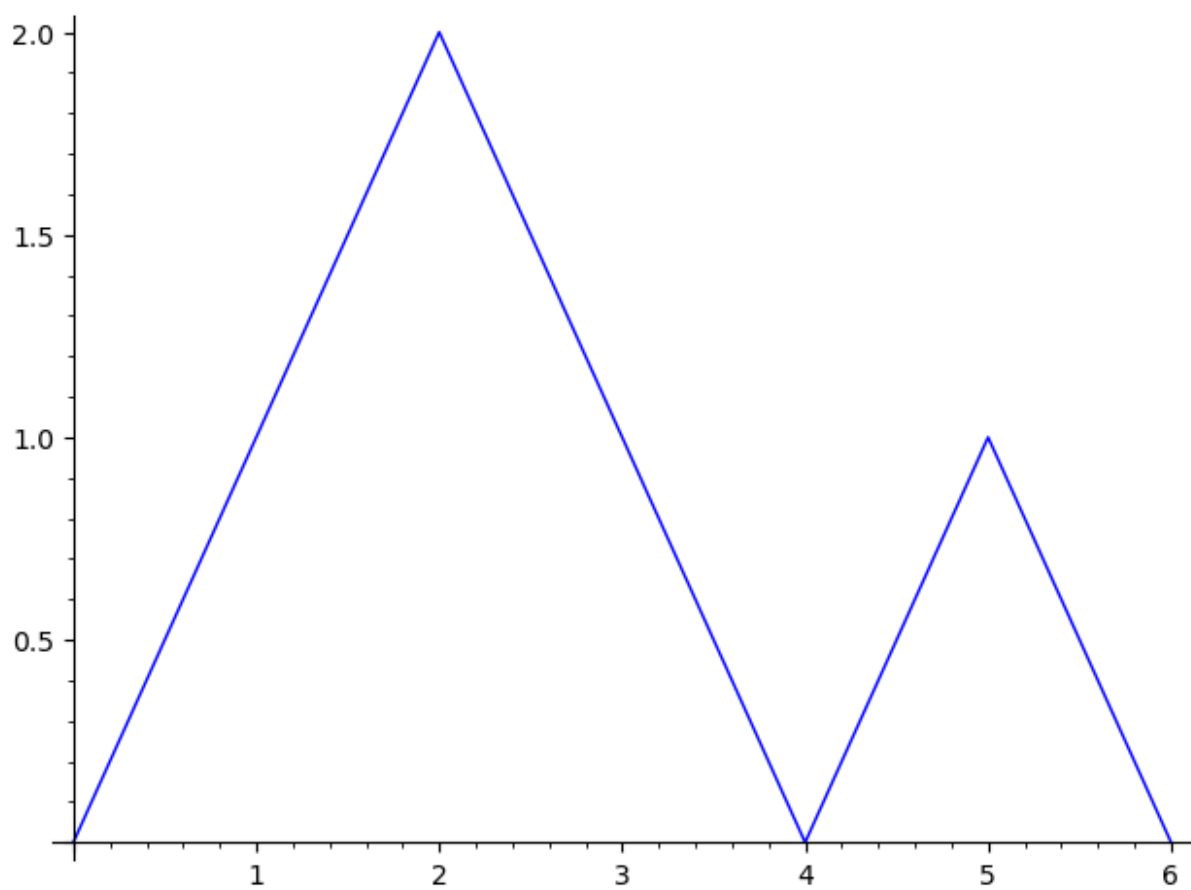
()()()



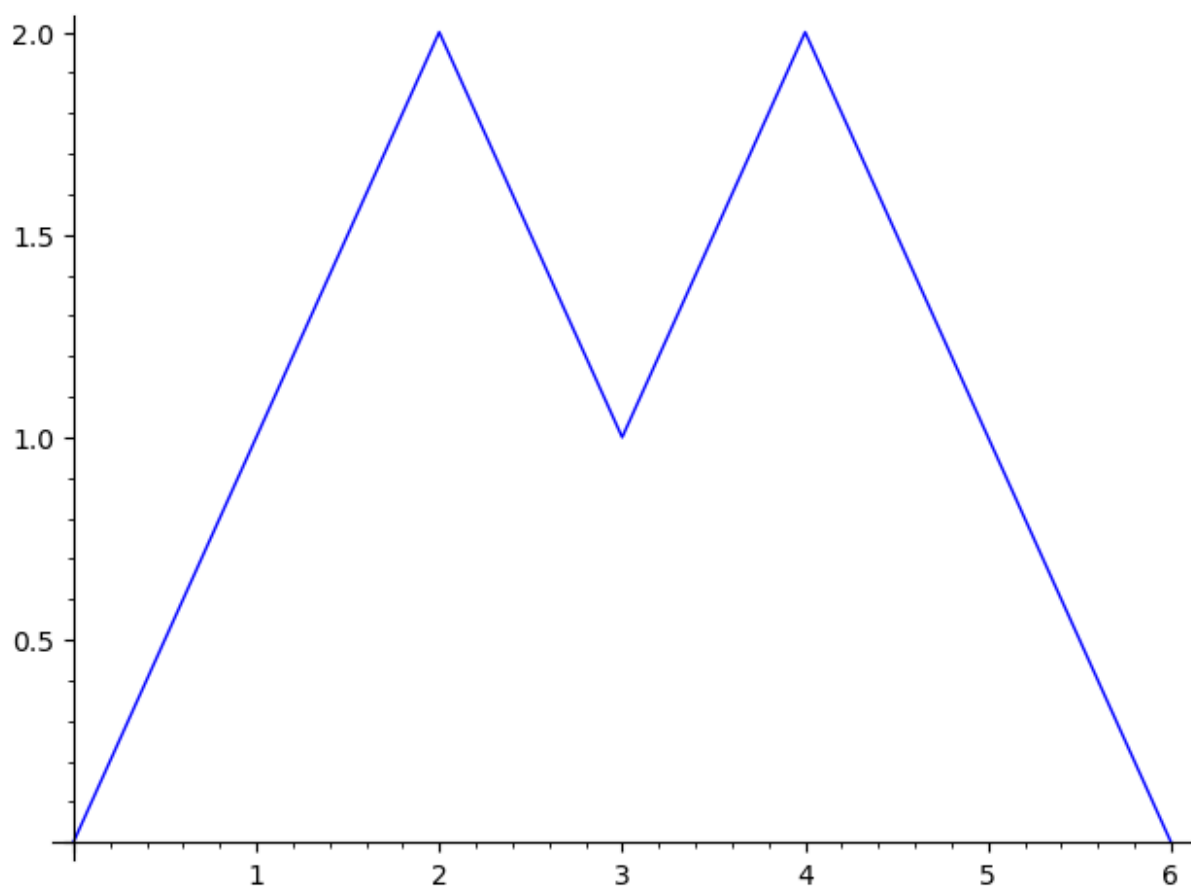
()(())



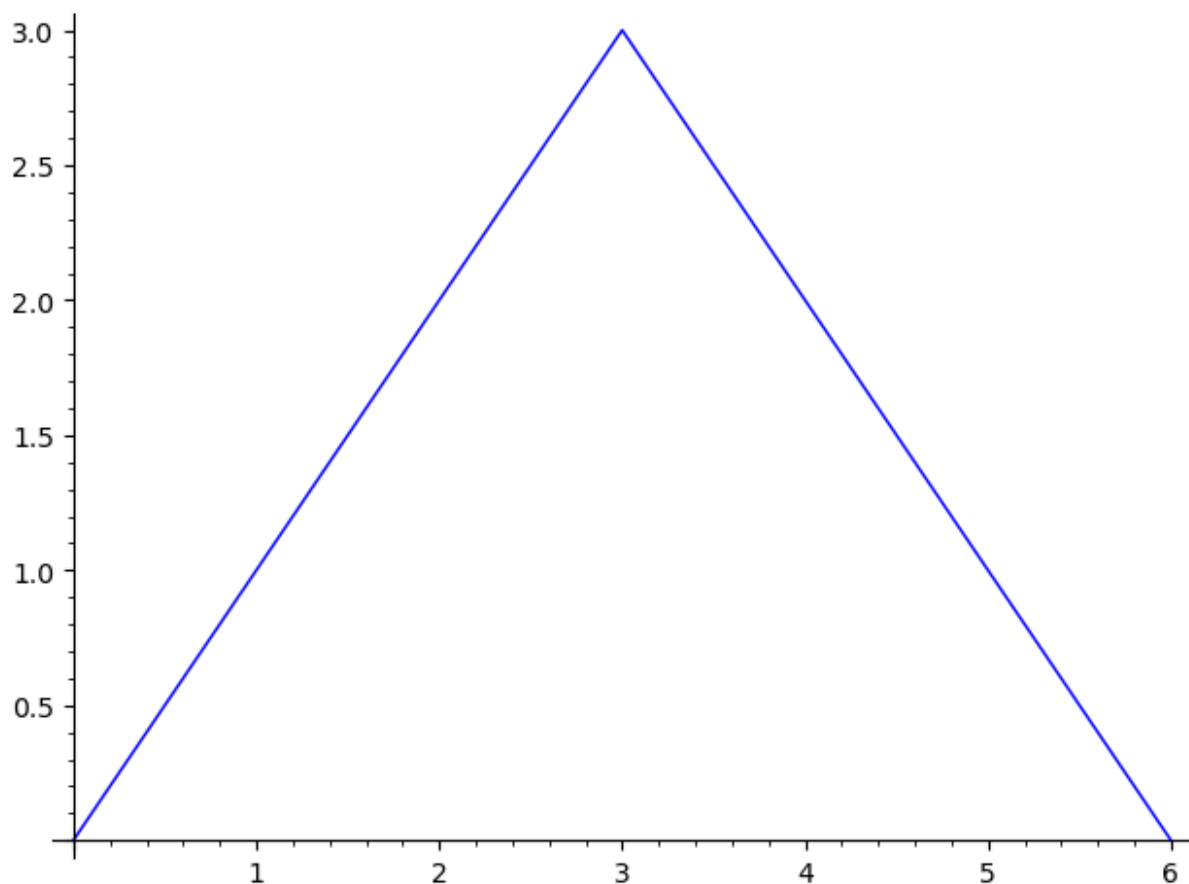
(())()



() ()



((()))



Below we define a function for the partial order by diagram containment.

Note how functions are structured in Python.

A function starts with 'def' and then has the name of the function followed by the inputs inside () and a colon : at the end of the first line.

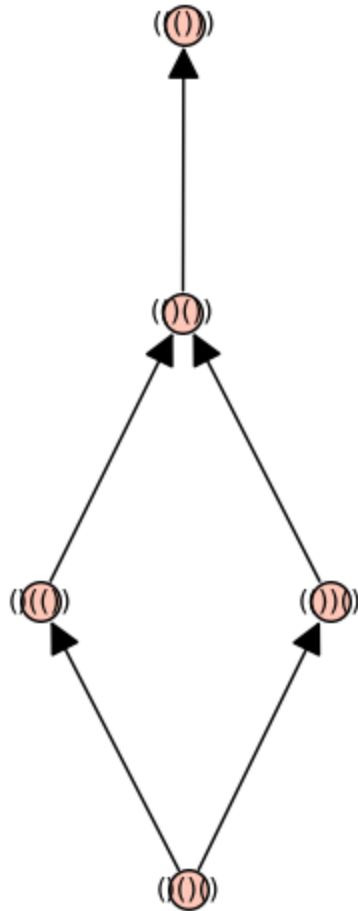
Then the next lines are indented and contain commands for the function to do. The last line includes 'return' to indicate what the function returns.

```
In [37]: def dyck_path_containment(x,y):  
         xpartition = x.to_partition()  
         ypartition = y.to_partition()  
         return xpartition.contains(ypartition)
```

Evaluate the block below to see the Dyck path poset for $n=3$.

```
In [38]: dyck_path_poset = Poset((DyckWords(3),dyck_path_containment))  
         dyck_path_poset.plot()
```


Out[38]:

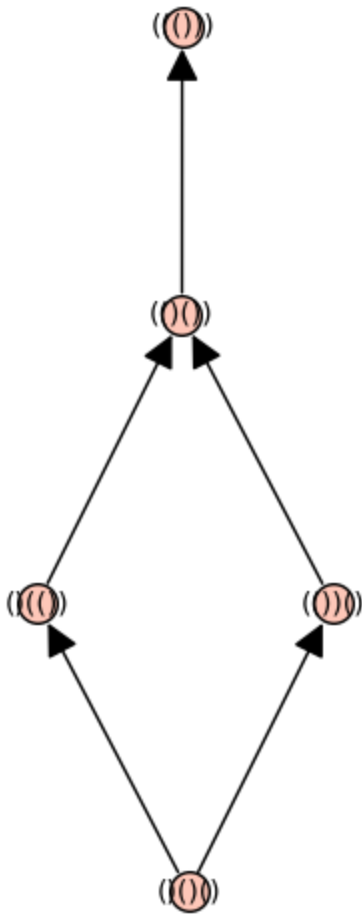


Copy the above commands to plot the Dyck path posets for $n=4$ and $n=5$. Give each poset a new name.

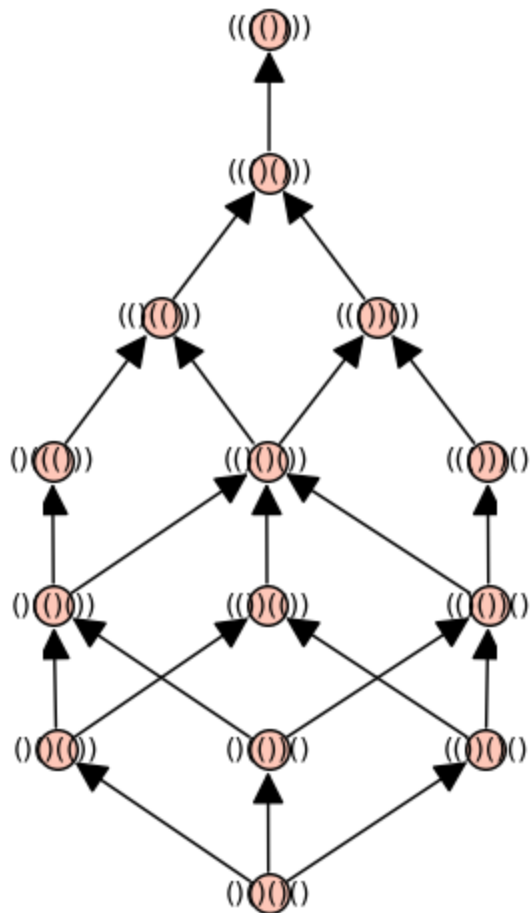
```
In [57]: DW = {
    "3": DyckWords(3),
    "4": DyckWords(4),
    "5": DyckWords(5)
}

DW_P = {}
for key, D in DW.items():
    DW_P[key] = Poset((DW[key], dyck_path_containment))
    print(f"Dyck path poset with {key} pairs of parantheses:")
    DW_P[key].plot().show()
print(DW_P)
```

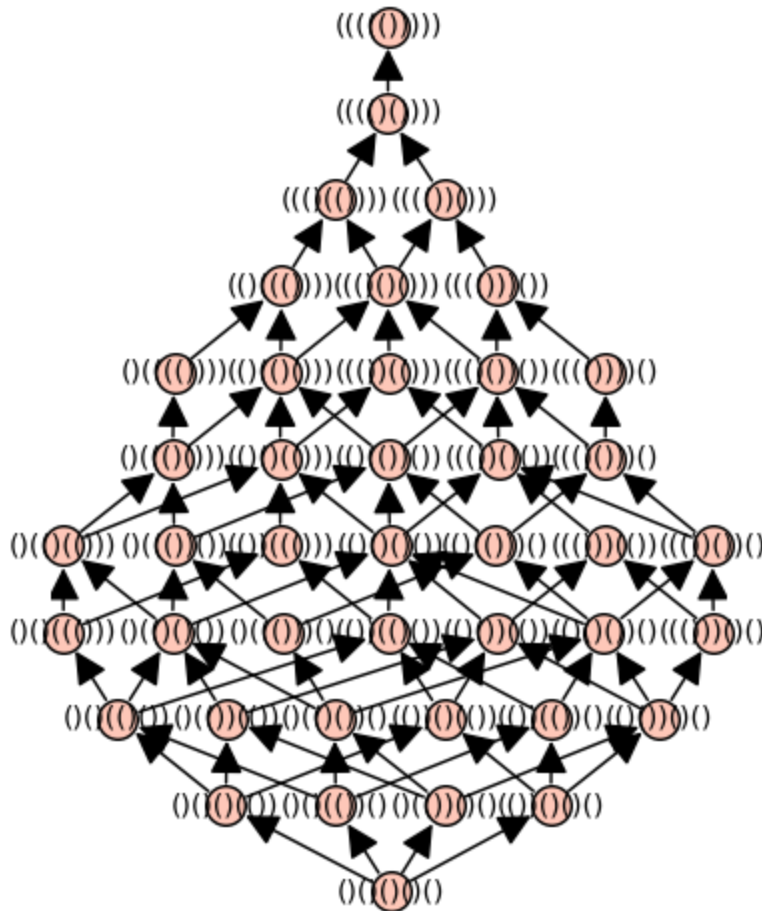
Dyck path poset with 3 pairs of parantheses:



Dyck path poset with 4 pairs of parantheses:



Dyck path poset with 5 pairs of parantheses:



{'3': Finite poset containing 5 elements, '4': Finite poset containing 14 elements, '5': Finite poset containing 42 elements}

Exercise 3: Evaluate the blocks and follow the directions below to explore lattice properties of the Dyck path poset.

We can ask Sage if the Dyck path poset is a lattice

```
In [43]: dyck_path_poset.is_lattice()
```

```
Out[43]: True
```

Sage has a different class for lattices. We can change our poset to that class and then ask whether it is distributive.

```
In [45]: #type something
dyck_path_lattice = LatticePoset(dyck_path_poset)
dyck_path_poset.is_lattice()
```

```
Out[45]: True
```

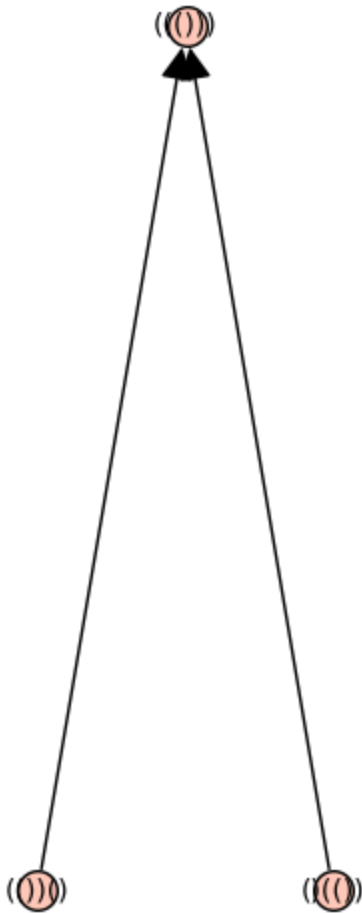
Put your cursor in the block below and press tab to find a function to help you tell if this lattice is distributive. Use this function.

```
In [46]: dyck_path_lattice.is_distributive()
```

```
Out[46]: True
```

Now use the command 'join_irreducibles_poset' on `dyck_path_lattice`. Plot the resulting poset and compare to what we discussed at the end of class yesterday.

```
In [53]: DW_P_JI = dyck_path_lattice.join_irreducibles_poset()  
DW_P_JI.plot().show()
```



Do the above sequence of commands for $n=4$ and $n=5$ to plot the posets of join irreducibles in those cases. What do you conclude?

```
In [58]: DW = {  
    "3": DyckWords(3),  
    "4": DyckWords(4),  
    "5": DyckWords(5)
```

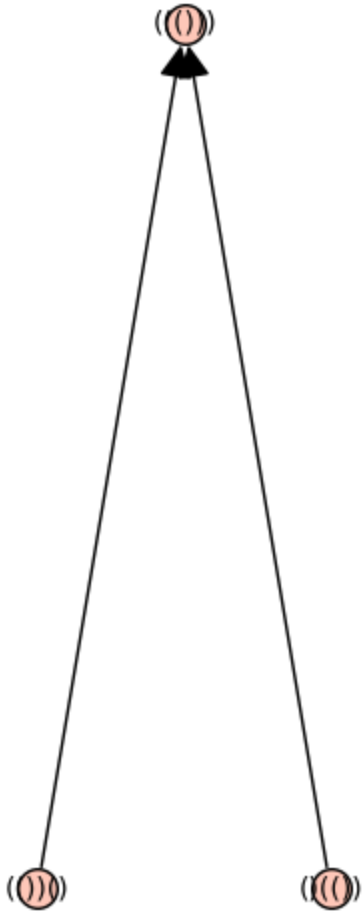
```

}

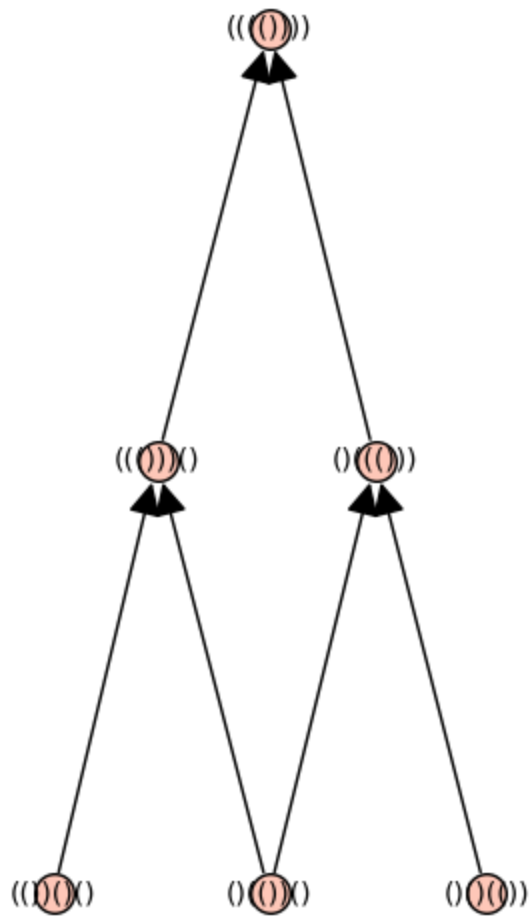
DW_LP = {}
for key,D in DW_P.items():
    DW_LP[key]= LatticePoset(DW_P[key]).join_irreducibles_poset()
    print(f"Join Irreducible Dyck path poset with {key} pairs of parantheses")
    DW_LP[key].plot().show()
print(DW_P)

```

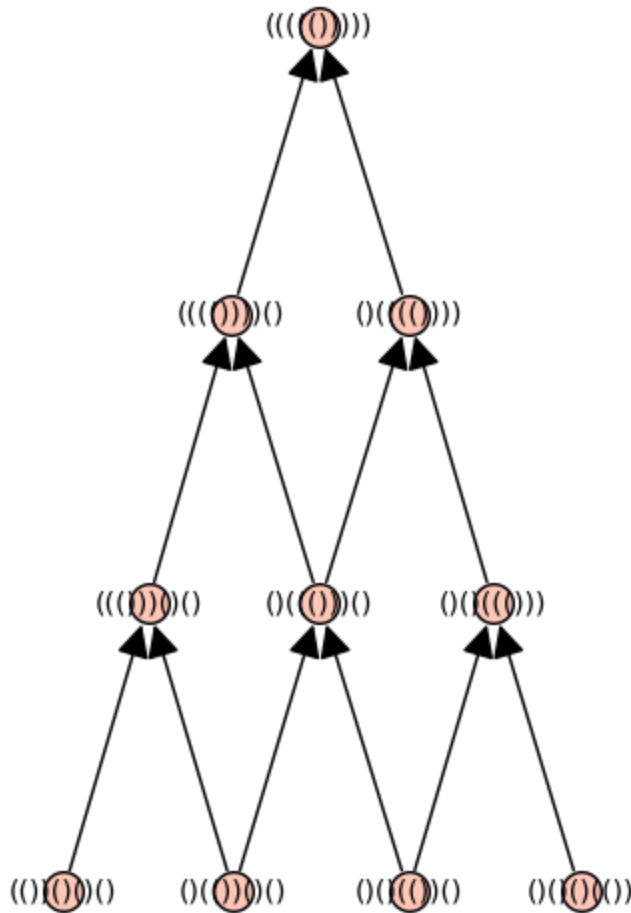
Join Irreducible Dyck path poset with 3 pairs of parantheses:



Join Irreducible Dyck path poset with 4 pairs of parantheses:



Join Irreducible Dyck path poset with 5 pairs of parantheses:



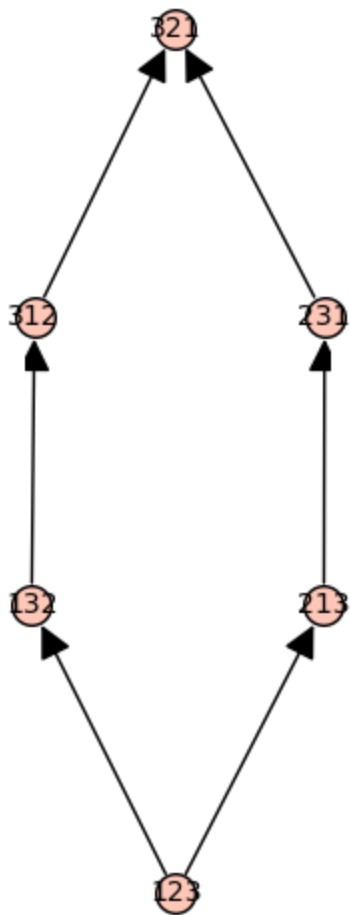
{'3': Finite poset containing 5 elements, '4': Finite poset containing 14 elements, '5': Finite poset containing 42 elements}

Exercise 4: Explore functions on Permutations (by typing `.` and `tab` after the function in the block below) to find and plot the strong and weak Bruhat order for $n=3,4,5$. Then check whether they are lattices, and if so, whether they are distributive.

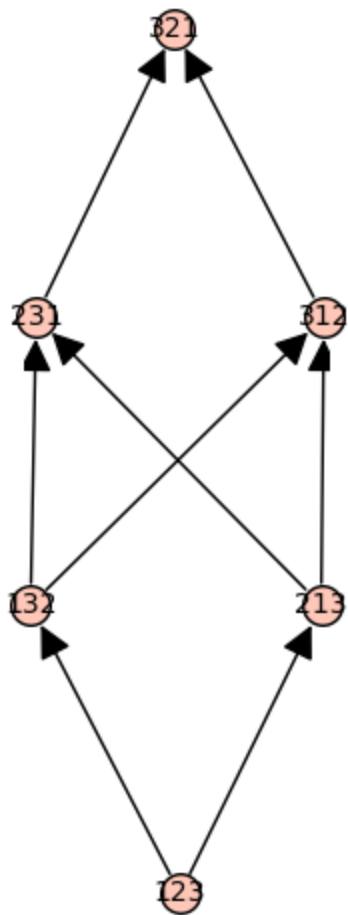
Hint: these posets are built-in, so you do not need to define the partial order function yourself.

```
In [78]: for n in [3,4,5]:      # as the worksheet says
          print(f"n={n}:")
          print(f"Bruhat weak order poset for permutations on [{n}]:")
          W = posets.SymmetricGroupWeakOrderPoset(n) # Bruhat weak order poset on S_n
          W.plot().show()    # strong Bruhat order on S_n
          print(f"Bruhat strong order poset for permutations on [{n}]:")
          B = posets.SymmetricGroupBruhatOrderPoset(n) #Bruhat strong order poset on S_n
          B.plot().show()    # strong Bruhat order on S_n
```

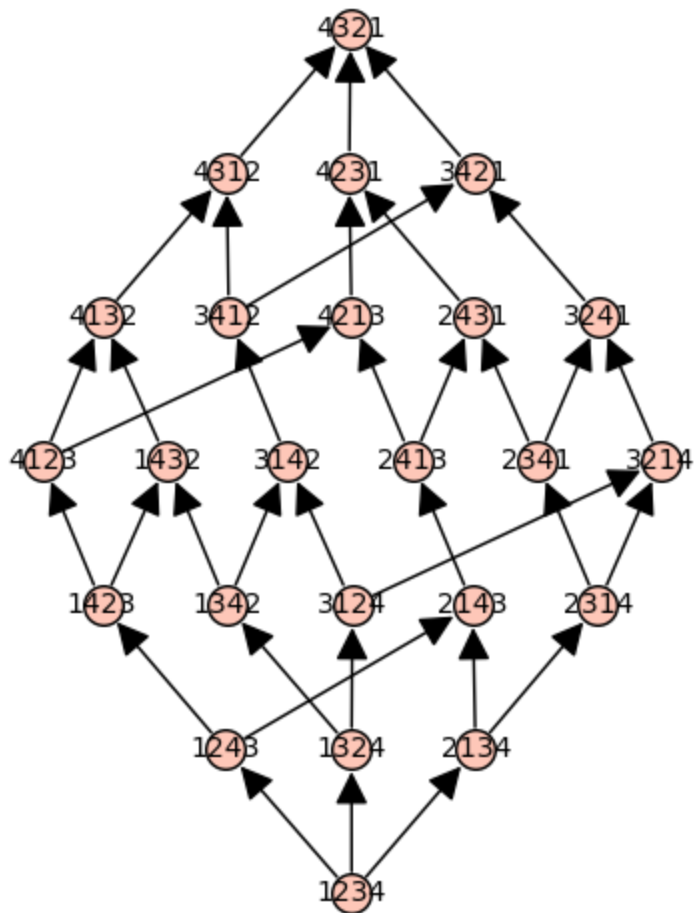
n=3:
Bruhat weak order poset for permutations on [3]:



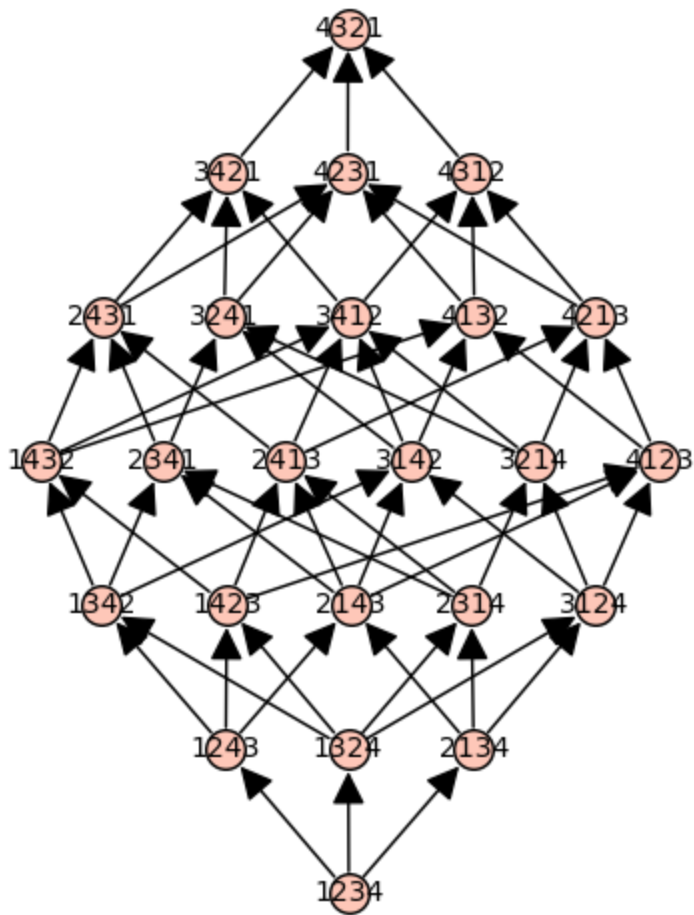
Bruhat strong order poset for permutations on [3]:



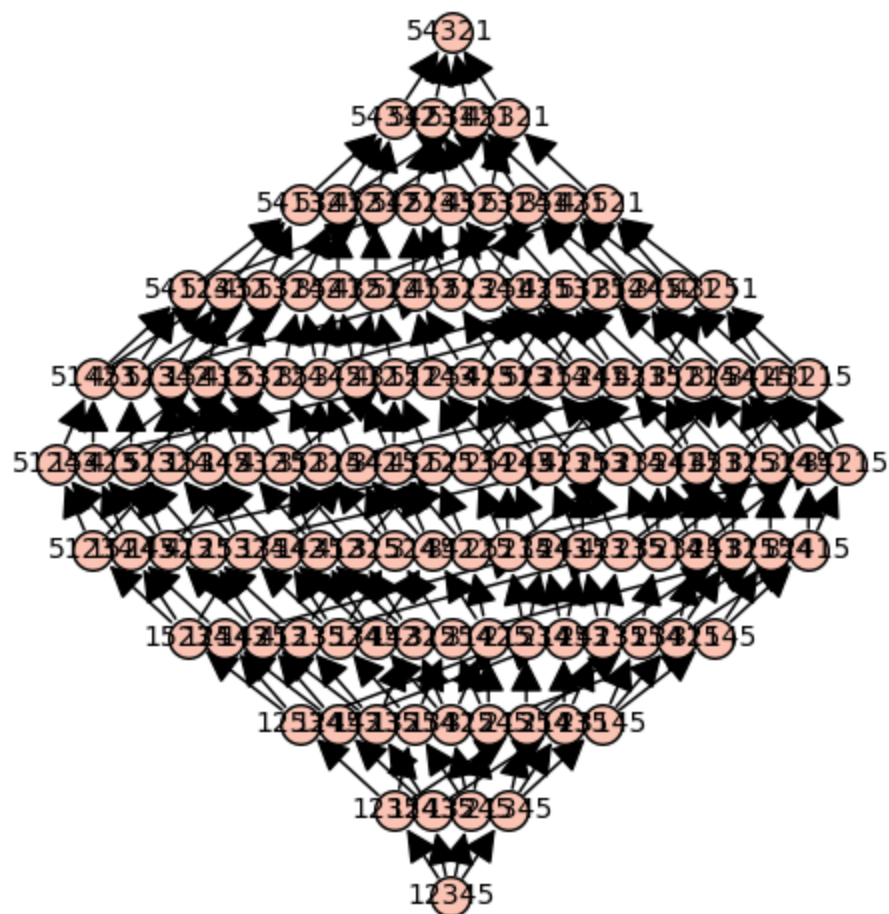
n=4:
 Bruhat weak order poset for permutations on [4]:



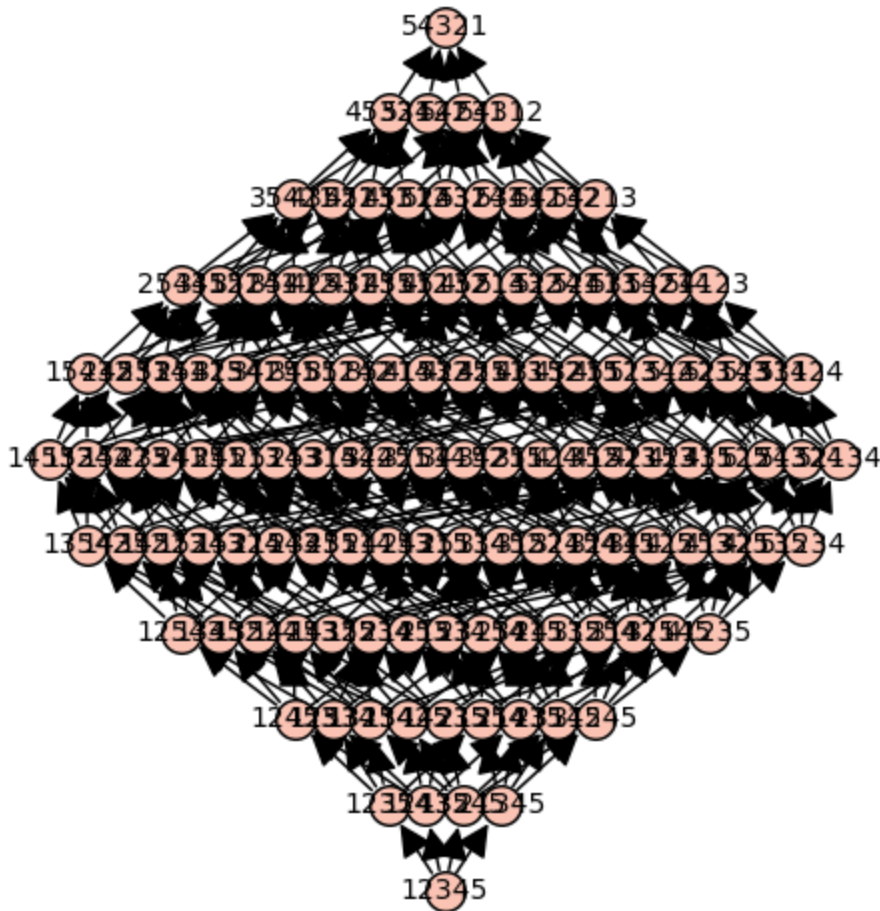
Bruhat strong order poset for permutations on [4]:



n=5:
 Bruhat weak order poset for permutations on [5]:



Bruhat strong order poset for permutations on [5]:



Exercise 5: Do some more exploration with posets in Sage. Write a sentence or two about what you discover.

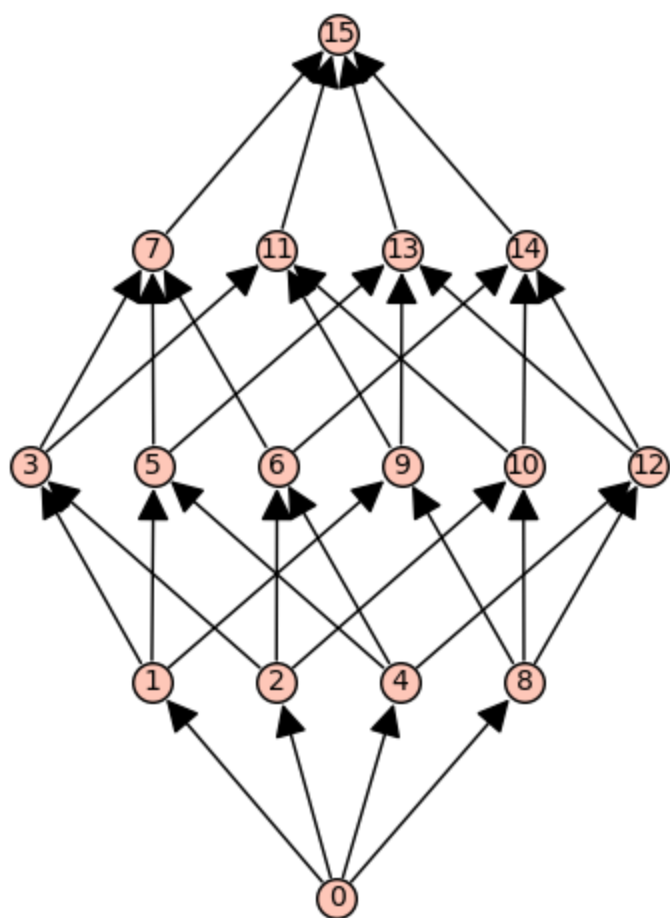
```
In [99]: n = 4
B = posets.BooleanLattice(n)
B.plot().show()

JI = B.join_irreducibles_poset()
JI.plot().show()
print("B size", B.cardinality(), "JI size", JI.cardinality())

m = 60
D = posets.DivisorLattice(m)
D.plot().show()

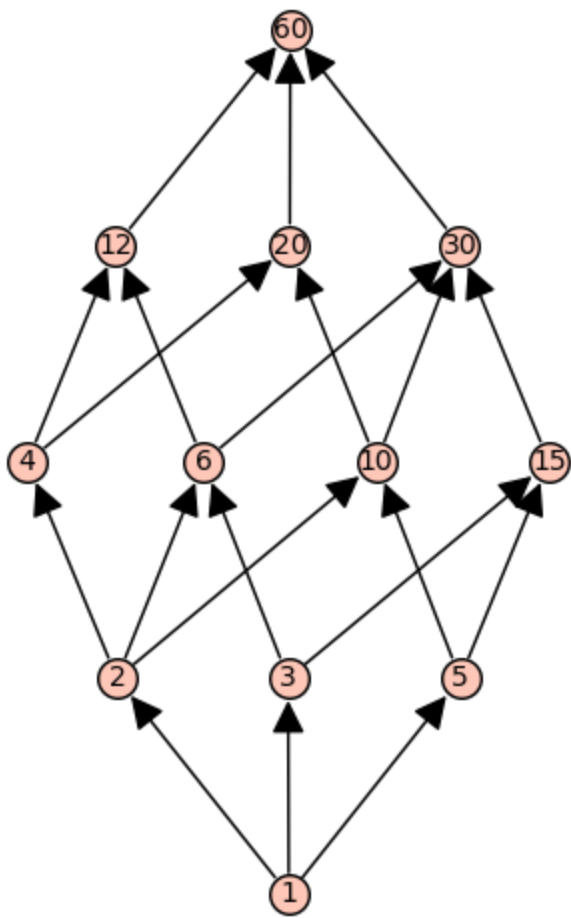
D.join_irreducibles_poset().plot().show()

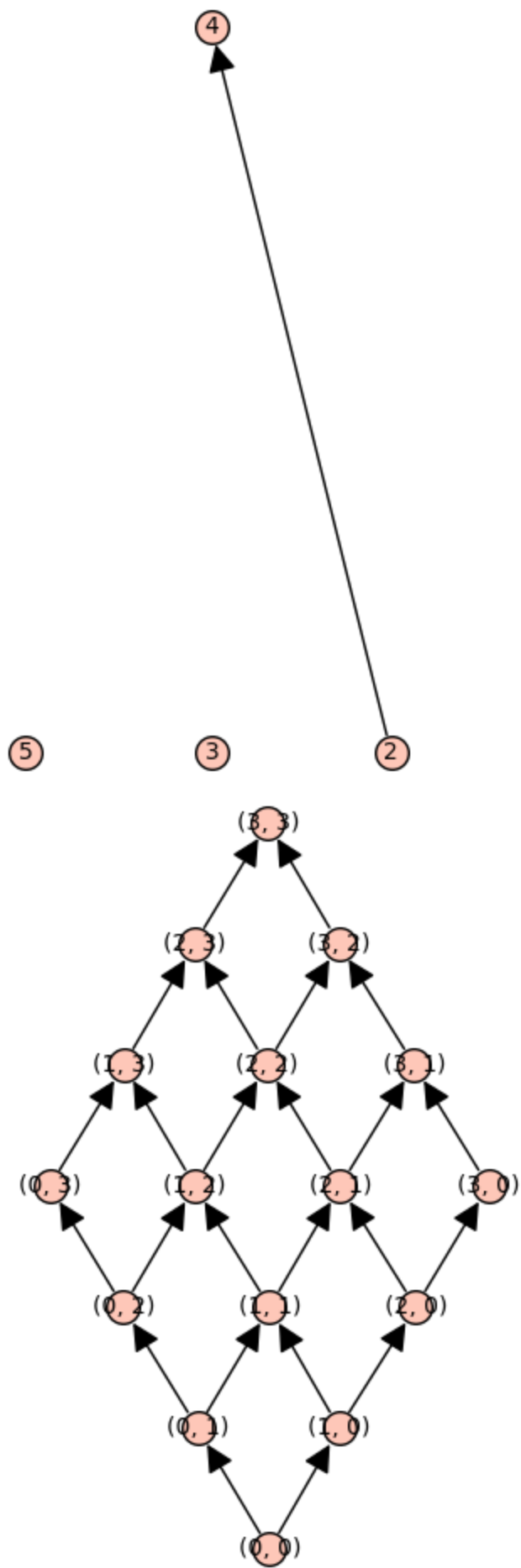
C = posets.ChainPoset(4)
G = C.product(C) # 4x4 grid lattice
G.plot().show()
G.join_irreducibles_poset().plot().show()
```

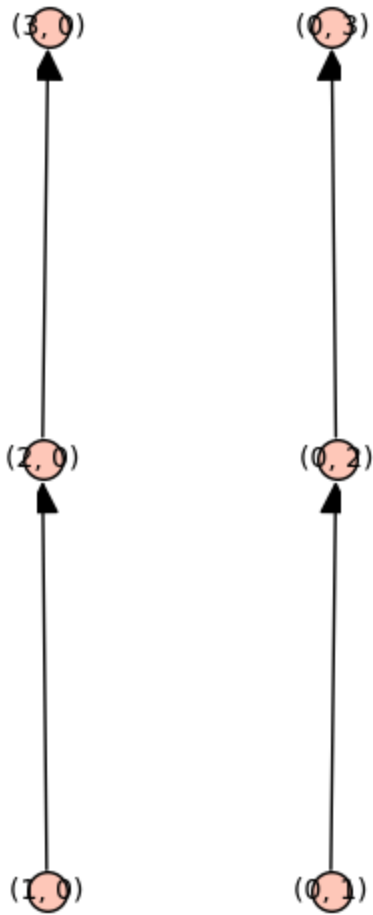


8 4 2 1

B size 16 JI size 4







There are a lot of useful built in posets and methods in sage. I could see this being useful for all kinds of things. Pretty cool.